

J17
HV+
2x
1x

F11
0461.500ER
R155
1k

HV+

HV-FUSED

R160
R161
R162
R163
R164
200k
200k
200k
200k
200k

D72
1.5SMC300CA
D73
1.5SMC300CA

Transient Voltage Suppression
Clamps transient spikes in TS DC-Link Voltage,
preventing damage to downstream components

12V Zener clamps gate voltage of Q20 to 12V,
and prevents noise or low voltage
accidentally triggering "HV Detected"
signal until 12V gate voltage.

D76
MSMZ4699TIG
12V zener

33k
R165

D77
E3-08
MSMZ4699-E3-08

J18
HV-
2x
1x

HV-

Q20
IPD80R4K5P7

200k
R168
10k
R174

1mA

U32
AZ431LBNTR-G1

Precision comparator
monitors voltage on Ref pin
compared to fixed 1.24V
reference.
When threshold met, current
flowthrough cathode to
GND, turning on the LED

U4
TLP475A

1
2
3
4
5
6

+GLV
+GLV
+GLV
+GLV

R185
1k
R188
1k
R184
47k
R189
10k

Q21
XP262N7002IR-G
Q22
XP262N7002IR-G

GND
GND
GND
GND

0.2mA

HV DETCTED
HV DETCTED

F16
150ma

PWR_FLAG
PWR_FLAG

J24
Conn_01x04

TS LV

Galvanic isolation between
Traction System (TS) and Low Voltage (LV) planes

4mm Creepage Distance on PCB
with conformal coating

SCS:
- Green LED turns on when HV not detected.
Safe state = off, so Green LED uses
HV_DETCTED signal
(short to GND means off = safe state)

MOSFETs are used to
amplify + invert signal

3.3v Zener Diode D77 protect inputs
to U13, clamping voltage to 3.3v

J21 Conn_01x01

4 HV_DETTECTED High Voltage Not Detected Signal

D68 D_TVS

R1 10k Pull down to ensure SCS Wire disconnect = short to GND GREEN LED off - Safe state

GND GND

Time constant 1ms rise, 2ms fall (through pull-down resistor). Asymmetry not ideal, but not a problem because the signals are slow.

J19 Conn_01x02

1 INV_CLOSE_PCHG_RLY

2 INV_CLOSE_MAIN_CONTACTOR

D71 D_TVS

D70 D_TVS

R158 10k

R159 10k

R166 10k

R167 10k

C82 100n

C87 100n

GND GND GND GND GND GND

J20 Conn_01x03

1 TS_READY Signal to VCU when TS able to start

2 TS_ON_INPUT Signal from VCU when driver initiates TSON

3 GREEN_LED_ON TSAL GREEN Signal to TSAL Logic PCB

D67 D_TVS

D69 D_TVS

D70 D_TVS

GND GND GND

J22 Conn_01x03

1 LOW_AIR_MECH_ON_IN

2 HIGH_AIR_MECH_ON_IN

3 PRCHG_MECH_ON_IN

+12V

R173 270

+12V

R176 270

+12V

R177 270

R180 10k

R181 10k

R182 10k

LOW_AIR_MECH_ON

HIGH_AIR_MECH_ON

PRCHG_MECH_ON

C89 100n

C90 100n

C91 100n

GND GND GND

J23 Conn_01x01

1 SHUTDOWN

SHUTDOWN

D86 SS14+

0.5A

F14 0448.500MR

+12V

PWR_FLAG

D88 D_TVS

C92 100u

GND GND

Diode prevents capacitors from powering the shutdown loop when the shutdown loop opens, without it the capacitors on the 12V supply of this board would keep the AIRs on for a short time after the loop opens

Upplups to provide auxiliary contacts a 'wetting' current' to prevent issues with contact oxidation in the long term

Auxiliary contacts (AIRs and precharge)

+GLV

F15 0448.500MR

+GLV_FUSED

PWR_FLAG

C93 100u

GND

Rules:

- The TS should never be enabled if TS_ON is low
- The TS should only be enabled after a rising edge of the TS_ON input. Whenever the shutdown circuit is opened, the TS should remain off until a rising edge is detected.

Goes high 0.1s after TS_ON goes high. Goes low immediately after TS_ON goes low. Rising edge here turns latch on, unless it's held in reset. Delay required to ensure rising edge happens after reset goes low.

Low when TS_ON is high. High when TS_ON is low.

Goes low 0.47s after shutdown circuit closes. Goes high 0.01s after shutdown circuit opens.

For this to go high, the following must be true:

- TS_ON is high (else the latch will be in reset).
- The shutdown loop has been closed for at least 0.47s (else the latch will be in reset).
- A rising edge on TS_ON has been detected at least 0.47s after the shutdown circuit was last open.

This last point guarantees rule EV4.11.4 (FSG 2026)

Close the +ve AIR when the TS is enabled and the inverter commands main contactor closed

Close the -ve AIR when the TS is enabled

Close the pre-charge relay when the TS is enabled and the inverter commands it closed

Vehicle: STAG 12
 Drawn By: Adam El
 Checked By: Ray W
 CAD Part: N/A
SUFST
 Sheet: TSAL GREEN
 File: TSAL_GREEN_A
Title: TSAL GR
 Size: A2